Example: prove $A = \{0^n 1^n \mid n \geq 0\}$ is not regular.

Assume A is regular.

Suppose pumping length of language A is p.

(Remember this is a fancy way of saying the DFA that recognize A has p states)

Now, the goal is to pick a string s in A of length at least p such that it violates the pumping lemma.

$S = 0^p 1^p \in A \qquad |S| = 2p$

$$S = \underbrace{0000\ldots\ldots 0}_{p\#\ 0's}\underbrace{1111\ldots\ldots 1}_{p\#\ 1's}$$

$S = xyz$

$$S = \underbrace{00000}_{x}\underbrace{\ldots 01111}_{y}\underbrace{\ldots 1}_{z}$$

# of ways I can divide s into 3 consecutive substrings is $\binom{2p}{2}$

However, by condition (3) of P.L, $|xy| \leq p$. so whatever, the decomposition for $s = xyz$, y could only consist of 0's.

Since, in any decomposition y can only consist of 0's, then xy^2z cannot be part of language a.

$$S = 0^p 1^p \qquad x = 0^a \quad y = 0^b \quad z = 0^c 1^p$$

$$a+b+c = p \qquad b > 0$$

$$xy^2z = 0^a 0^{2b} 0^c 1^p = 0^{a+2b+c} 1^p$$

$$a+2b+c > p$$

$$\therefore xy^2z \notin A$$

Clearly, $S = 0^p 1^p$ violates pumping lemma for all possible decompositions and all p values.

$\therefore$ A cannot be regular.

How about picking $s = 0^{p/2} 1^{p/2}$ to show the contradiction?

Not a good string to pick.

Example 02: Let $\Sigma = \{0, 1, 2\}$

show that $B = \{0^n 1^n 2^n : n \geq 0\}$ is not regular.

Proof by contradiction.

Assume B is regular.

Then B must follow pumping lemma.

Suppose pumping length is p.

Let us pick $s \in B$, s.t $|s| \geq p$.

$s = 0^p 1^p 2^p$, $|s| = 3p$ $\quad 3p \geq p$

By pumping lemma, S should be

able to be decomposed into $s = xyz$

s.t 1. $xy^iz \in B$, $\forall i \geq 0$

2. $|y| > 0$

3. $|xy| \leq p$.

By ③ of pumping lemma, for $S = 0^p 1^p 2^p$, in any decomposition y could only consist of 0's.
Also, by ② of pumping lemma y must consist at least one 0.

Hence, $xy^iz \notin B$, as when you pump y, the # of 0's would change.

This is a contradiction

$\therefore$ B is not regular

Example 03:

$\Sigma = \{0,1\}$

$E = \{ 0^i 1^j : i > j \}$

is not regular.

What would be a good string to pick?

$s = 0^{p+1} 1^p$

Proof: Assume E is regular.

Thus, E follows pumping lemma.

Pick $S = 0^{p+1} 1^{p}$, $s \in E$, $|s| = 2p+1 \geq p$

Since, E is regular and follows pumping lemma, s must be decomposible to $s = xyz$ s.t

1. $xy^{i}z \in E$, $\forall i \geq 0$
2. $|y| > 0$
3. $|xy| \leq p$.

By 2 and 3, In any decomposition of $S = 0^{p+1} 1^{p}$, y should only consist of 0's and it should at least consist one 0.

$\therefore$ if you consider $xy^{0}z$, it should contain either same number of 0's as 1's or less number of 0's compared to number of 1's.

Hence, $xy^{0}z \notin E$.

$xy^{0}z = xz$

$F = \{ ww \mid w \in \{0,1\}^* \}$ show that F is not regular.

Question: what if we pick $s = 0^p 0^p$ as the string s?

you cannot use this string for this language.
since s is pumpable.

$s = 0^p 0^p$, let $a \in \mathbb{N}$ s.t a is a even number.

now $s = 0^{p-a} 0^a 0^p$

$$xy^iz = 0^{p-a} 0^{ai} 0^p$$

$$|xy^iz| = (p-a) + ai + p = 2p + a(i-1)$$

$2p + a(i-1)$ is an even number

$$0^{2p+a(i-1)} \in P$$

This not a good string to pick.

Proof: Assume F is regular. Thus, it must follow pumping lemma.

Suppose pumping length is p.

Let us pick $S = 0^p 1 0^p 1$, where $S \in F$ and $|S| \geq p$

$$|S| = 2p + 2$$

Then, S must be able to be decomposed into $S = xyz$ s.t

1. $xy^iz \in F$
2. $|y| > 0$
3. $|xy| \leq p$

So, by 3, in any decomposition of S, y must only contain 0's.

And By 2 y must at least contain one 0.

Then suppose, you allocate a # of 0's to x b # of 0's to y and rest of the string to z

$$S = \underbrace{0^a}_{x}\ \underbrace{0^b}_{y}\ \underbrace{0^{p-(a+b)}\ 1\ 0^p\ 1}_{z}$$

$$xy^iz = 0^a\ 0^{bi}\ 0^{p-(a+b)}\ 1\ 0^p\ 1$$

Then consider xy^0z

$$xy^0z = xz = 0^a\ 0^{p-(a+b)}\ 1\ 0^p\ 1$$

$$= 0^{p-b}\ 1\ 0^p\ 1$$

we know that $b > 0$ by 2

$$p-b < p$$

$$0^{p-b}\ 1\ 0^p\ 1 \notin F$$

Thus F is not regular.

$D = \{ 1^{n^2} \mid n \geq 0, n \in \mathbb{Z}^+ \}$, $\Sigma = \{1\}$

Prove that D is not regular.

$n = 0$ $\quad$ $n = 1$ $\quad$ $n = 2$ $\quad$ $n = 3$

ε $\quad$ $1^{1^2} = 1$ $\quad$ $1^{2^2} = 1111$ $\quad$ $1^{3^2} = 111111111$

$n = p$ $\quad$ $n = p+1$

1^{p^2} $\quad$ $1^{(p+1)^2}$

Assume D is regular.

Then D should follow pumping Lemma.

Suppose pumping length is p.

$S = 1^{p^2}$ $\quad |S| = p^2 \geq p$

So, S should be able to be decomposed into $S = xyz$ s.t

1. $xy^iz \in D, \forall i \geq 0$
2. $|y| > 0$
3. $|xy| \leq p$.

By ② and ③, in any decomposition of S, y should only consist of 1's and $|xy| \leq p$

Now, we consider all the possible decompositions at once.

suppose you allocate a # of 1's to x, and b # of 1's to y,

then $b > 0$ (by ②), and $a+b \le p$ (by ③)

$$s = xyz = \underbrace{1^{a}}_{x}\,\underbrace{1^{b}}_{y}\,\underbrace{1^{p-(a+b)}\;1^{p^2-p}}_{z}$$

consider xy^iz

$$xy^iz = 1^{a}\,1^{bi}\,1^{p^2-(a+b)}$$

$i=2$

$$|xy^2z| = \cancel{a} + \overset{b}{\cancel{2b}} + p^2 - (\cancel{a}+\cancel{b}) = p^2+b$$

$$p^2 \le p^2 \le p^2$$

$$p^2 < \underbrace{p^2+b}_{|xy^2z|} \le p^2+p \qquad (0 < b \le p)$$

$$\le p(p+1)$$

$$\le p(p+1) + (p+1) - (p+1)$$

$$\le (p+1)(p+1) - (p+1)$$

$$\le (p+1)^2 - (p+1) \qquad < (p+1)^2$$

$$p^2 < \underbrace{p^2+b}_{|xy^2z|} \le (p+1)^2 - (p+1) < (p+1)^2$$

$p^2 < |xy^2z| < (p+1)^2$

obviously, in order for xy^2z to be in D it needs to have the length at least $(p+1)^2$.

But we showed that this is not the case.

$\therefore$ $xy^2z \notin D$

$\therefore$ D is not regular.